

CS 180 Discussion #3

Filters & Frequencies!



Disc 2 Feedback Form



Konpat

Agenda!

Today we'll cover:

- Convolution + filters
- Frequency analysis

Logistics:

- HW 2 released!

Move to a person next to you,
And introduce yourself 😊 (1 min)

“Hi what’s your name?”

“What’s your favorite fruit?”

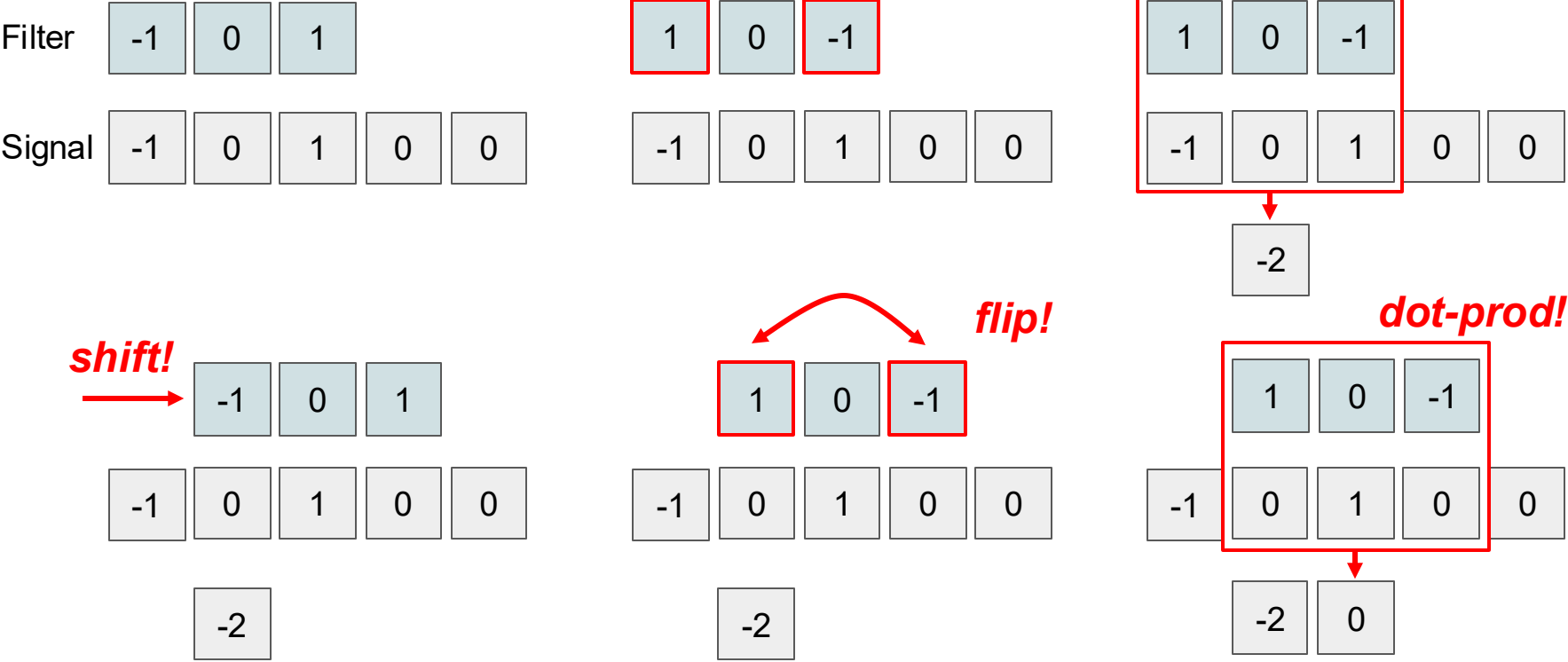
“Where are you heading to after this?”

Let’s work in pairs!

Convolutions

Convolution + filters

Visualizing the process:



*Output is smaller! How to fix?

Convolution + filters + padding

-1	0	1
----	---	---

0	-1	0	1	0	0	0
---	----	---	---	---	---	---

flip!

1	0	-1
---	---	----

0	-1	0	1	0	0	0
---	----	---	---	---	---	---

dot-prod!

1	0	-1				
0	-1	0	1	0	0	0

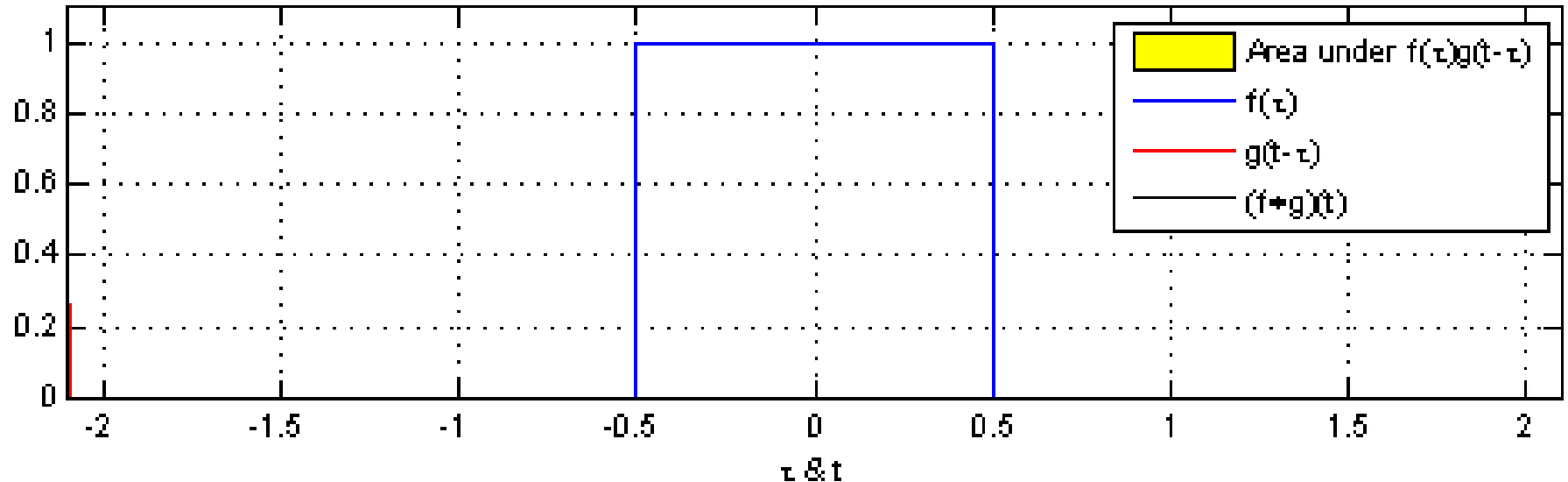
↓

0

*Same size
*So many kinds of padding

Convolution visualized, 1D

Convolve box filter (red), with another box filter (blue)



From <https://en.wikipedia.org/wiki/Convolution>

Convolution + filters

f = signal
g = kernel

$$(f * g)(t) := \int_{-\infty}^{\infty} f(\tau) \overbrace{g(t - \tau)}^{\text{Flip \& Shift}} d\tau$$

Memorize it like convention; "Flip, shift, dot product"

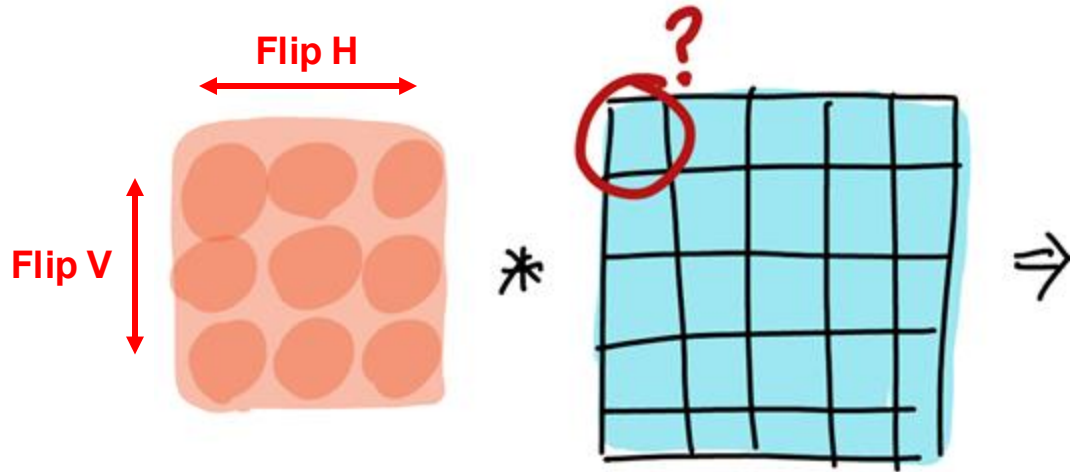
Commutative! $f * g = g * f$

From <https://en.wikipedia.org/wiki/Convolution>

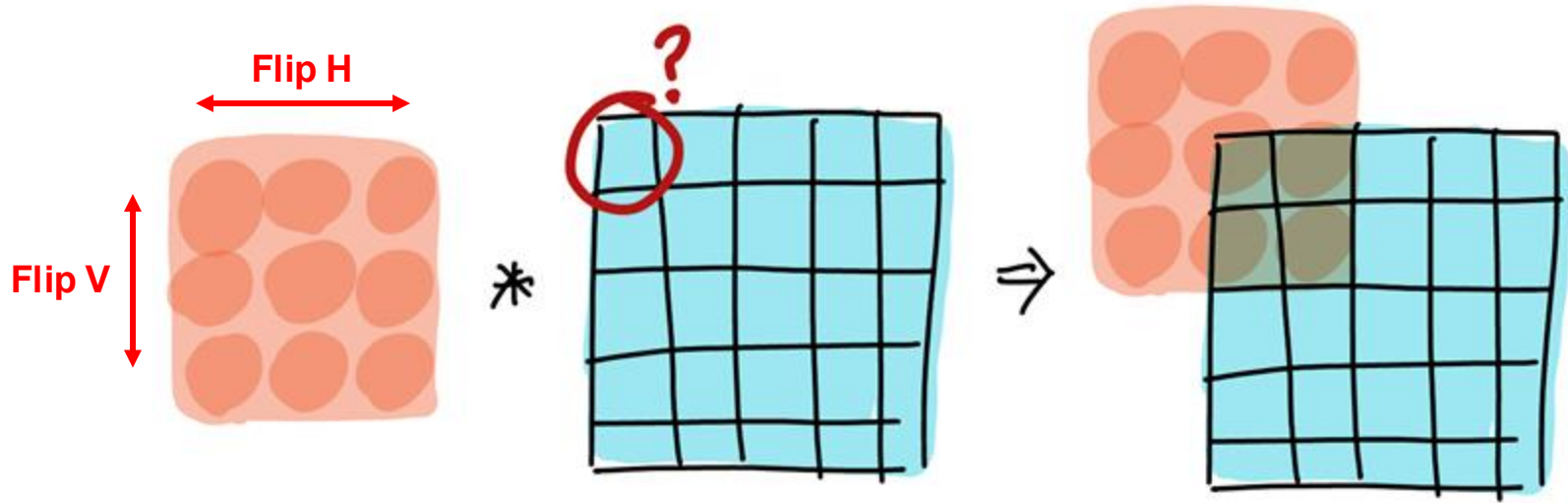
2D Convolutions

Convolution visualized, 2D

Sanity check #1: how do we align?

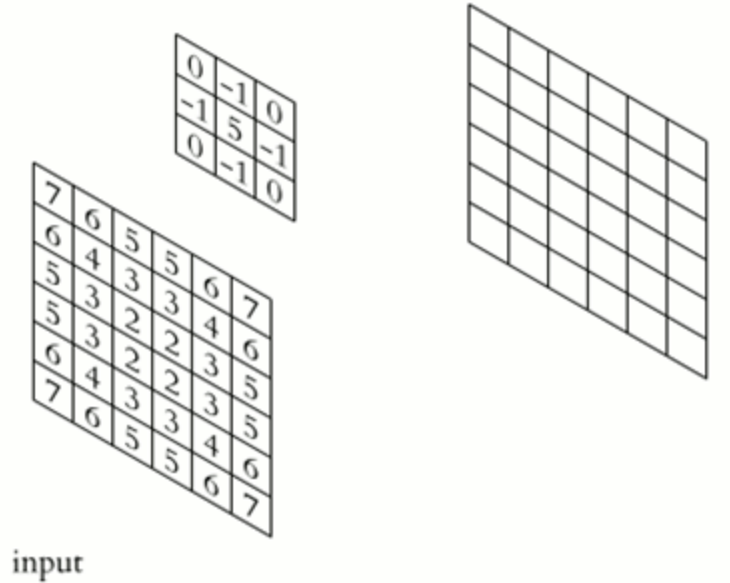


Convolution visualized, 2D



Sanity check #2: what do we do for missing values?

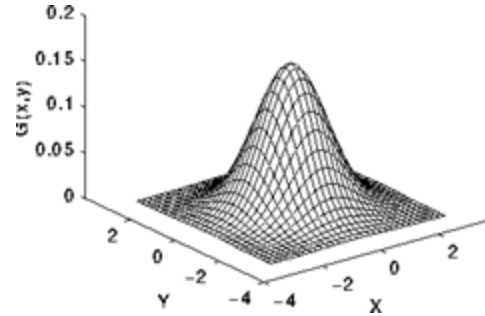
Convolution visualized, 2D



From <https://en.wikipedia.org/wiki/Convolution>

Discussion Q #1

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$



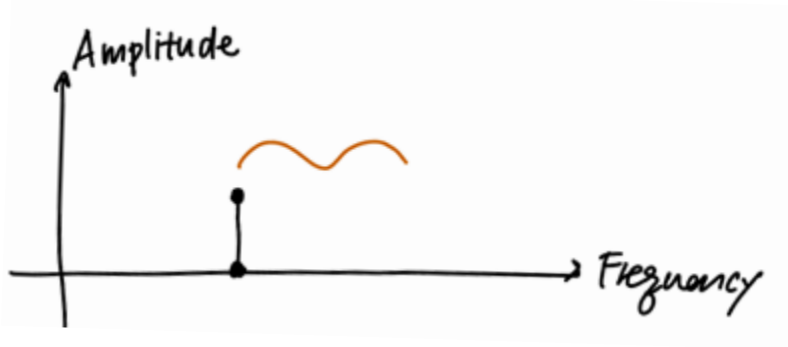
-1	0	+1
-2	0	+2
-1	0	+1

Gx

Frequencies!

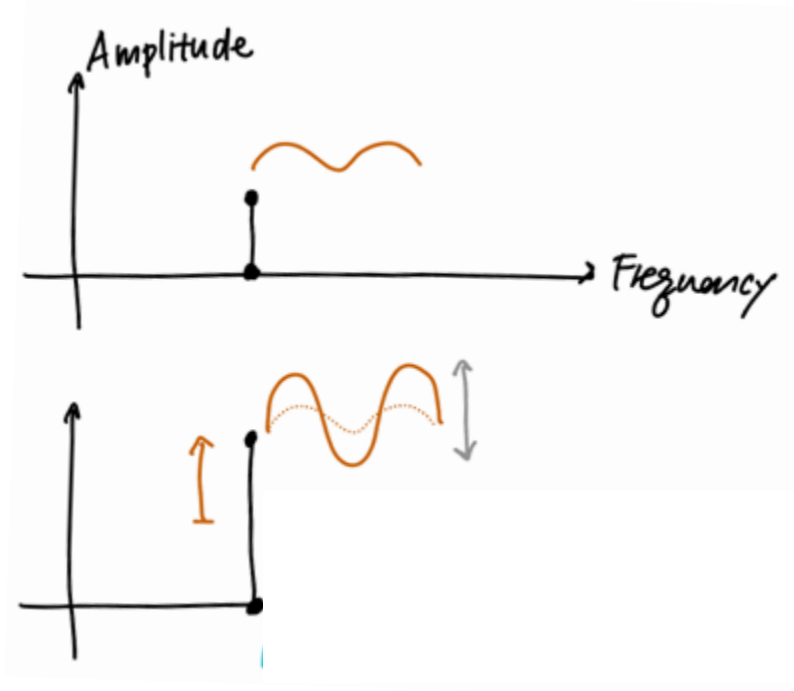
Studying frequencies (Fourier Analysis)

Visualize the **magnitude** ("how much") and **phase** ("shift")



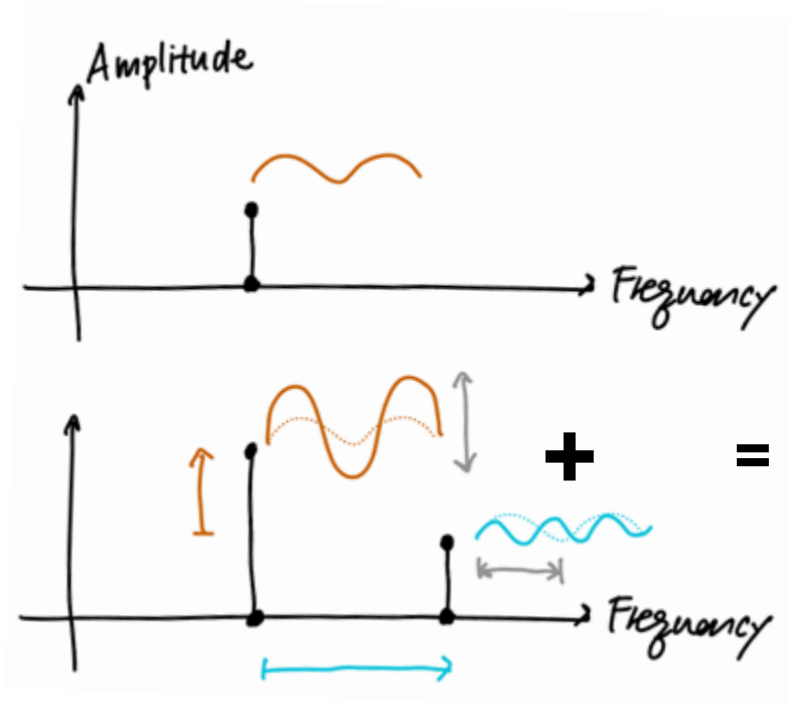
Studying frequencies (Fourier Analysis)

Visualize the **magnitude** ("how much") and **phase** ("shift")



Studying frequencies (Fourier Analysis)

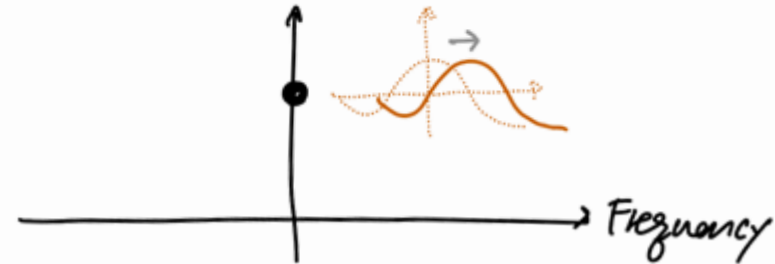
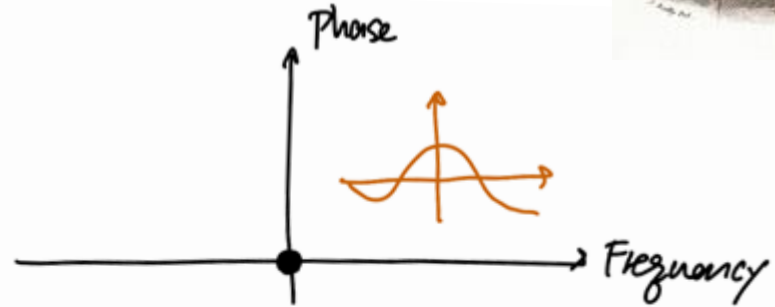
Visualize the **magnitude** ("how much") and **phase** ("shift")



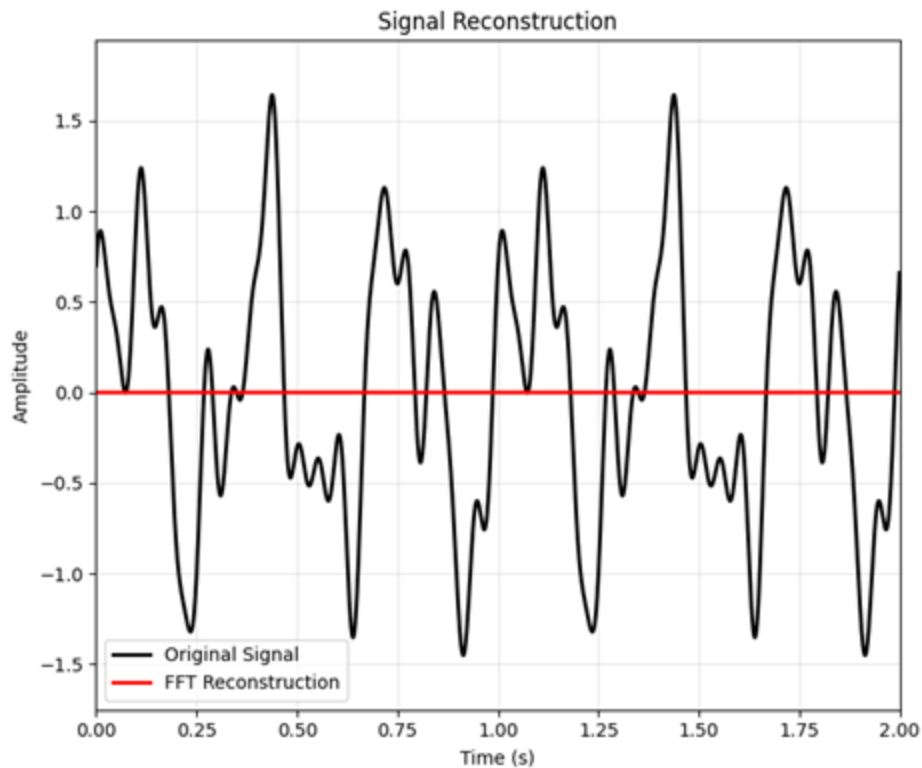
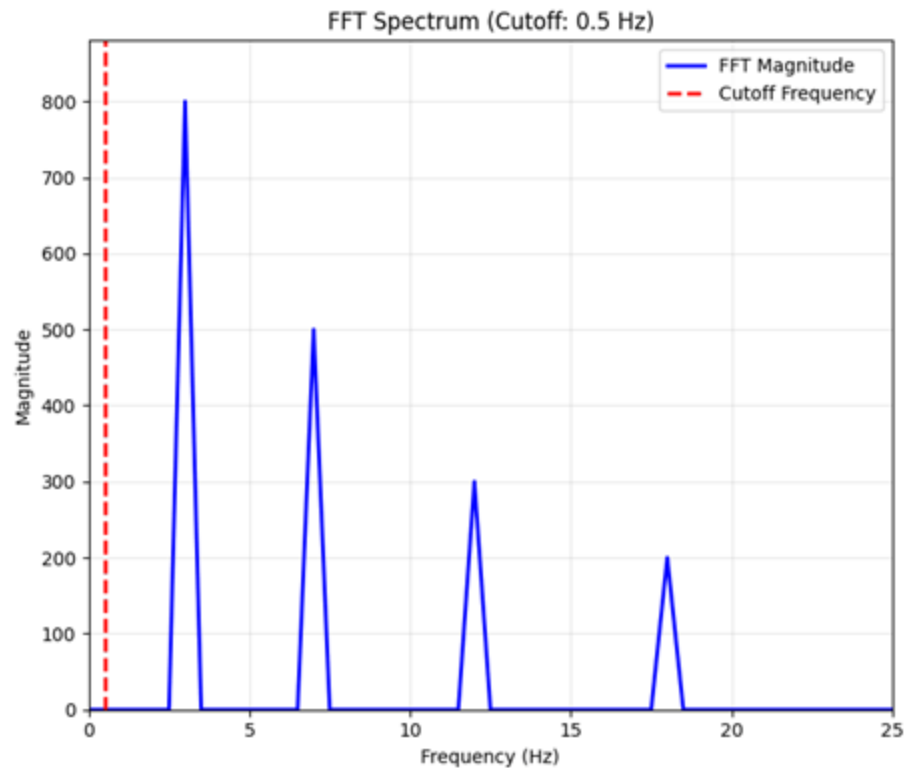
+ = complicated signal

Studying frequencies (Fourier Analysis)

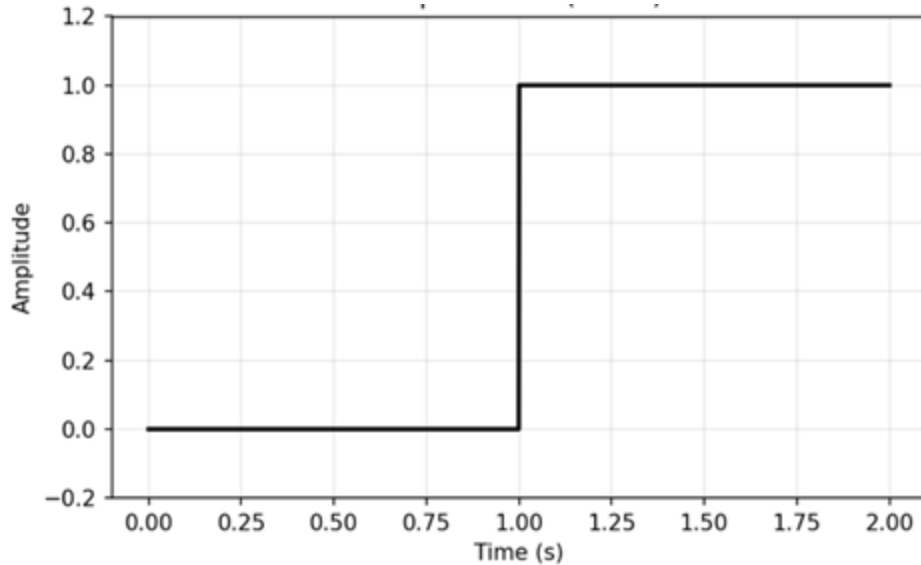
Visualize the **magnitude** ("how much") and **phase** ("shift")



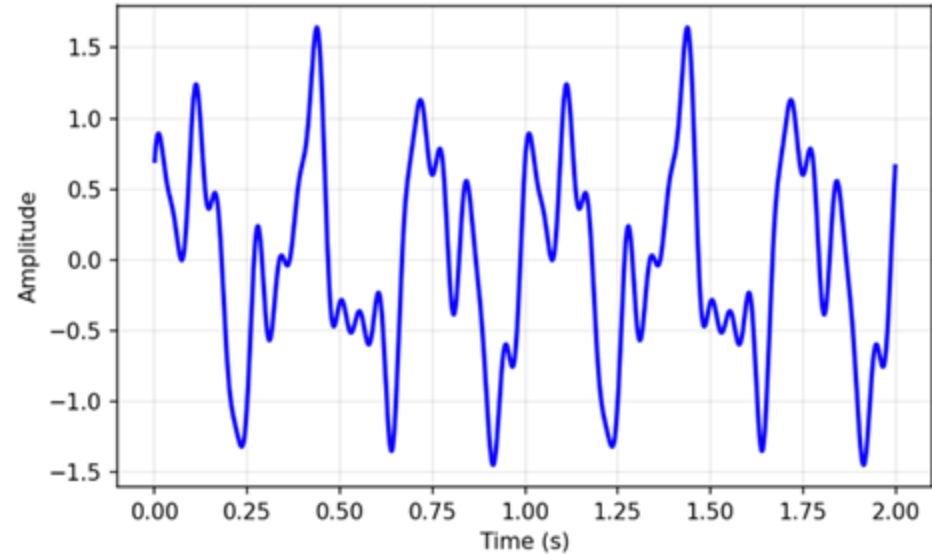
Reading FFT plots



Which is **easier** to represent in frequency space?



A



B

Edges are very high frequency

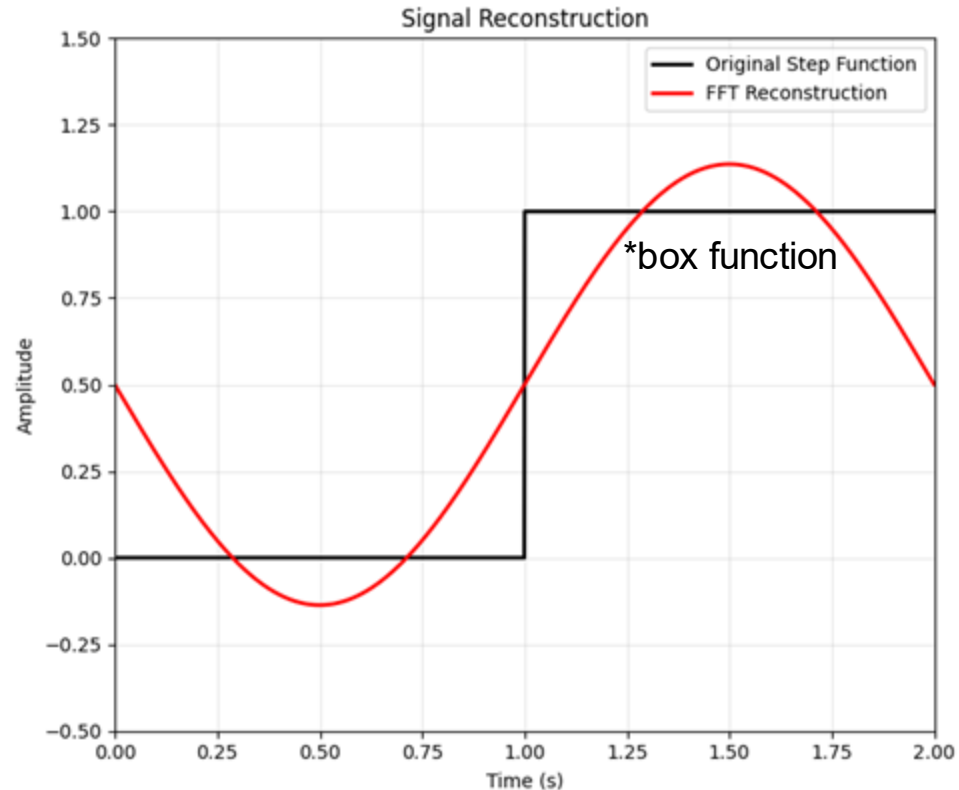
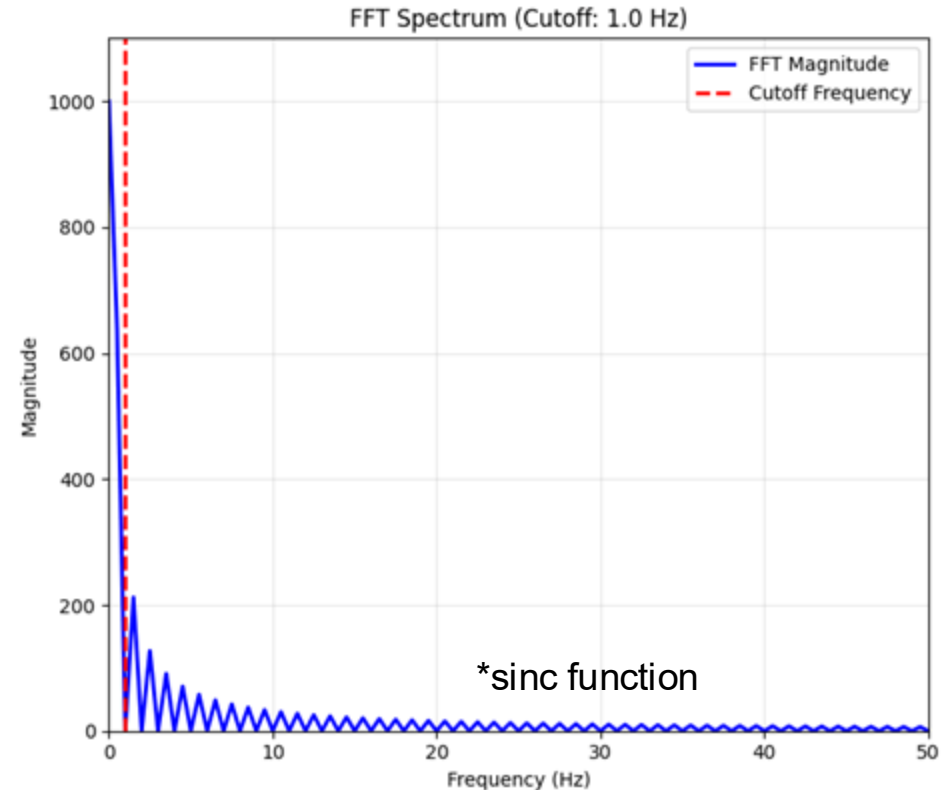
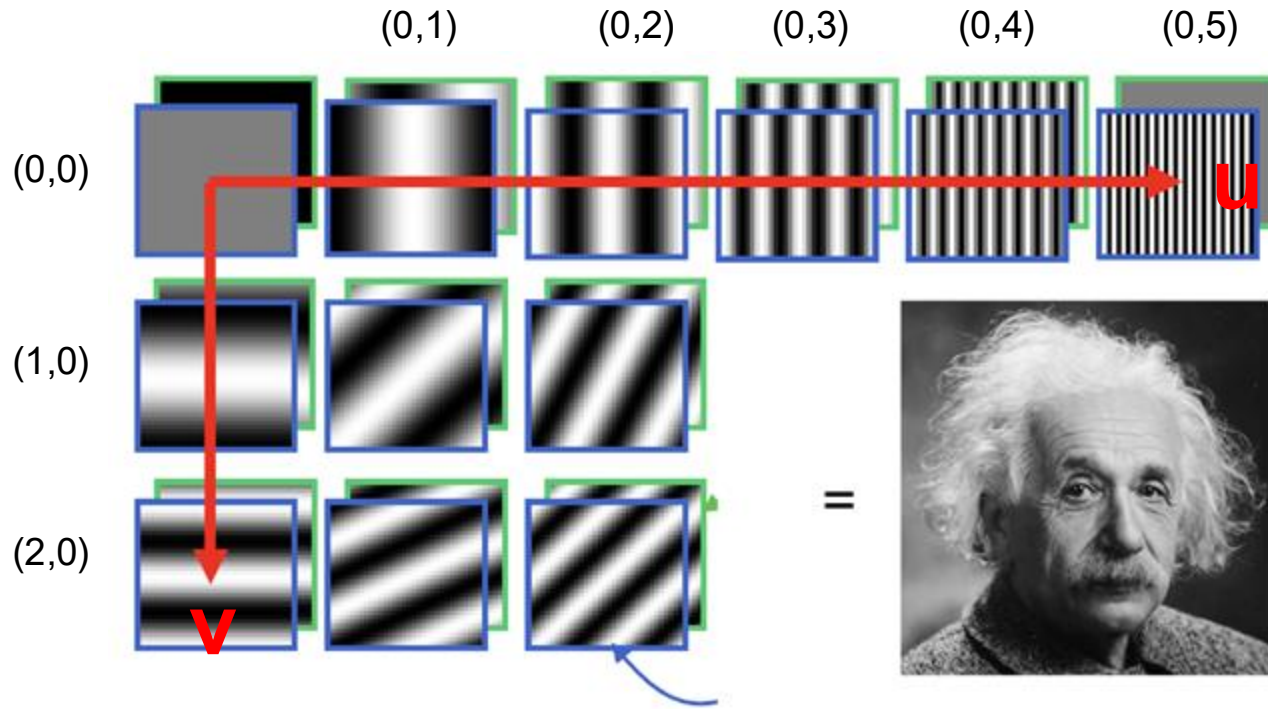


Image Frequencies!

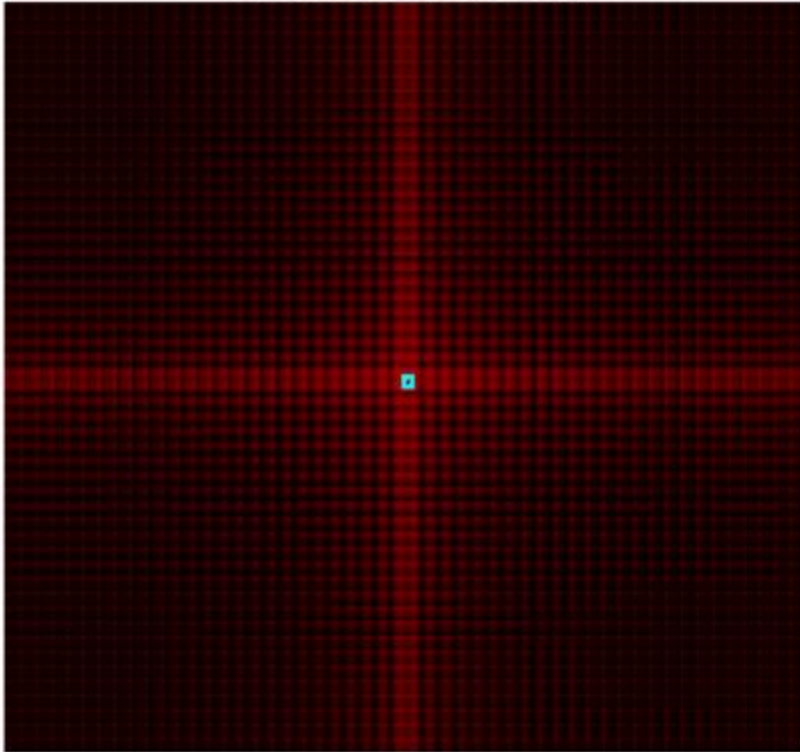
Frequency amplitudes map



*Same storage size. $H \times W = \text{num}(\text{amplitudes}) + \text{num}(\text{phase})!$

Step functions in an image

FFT with Mask (Size: 3px)



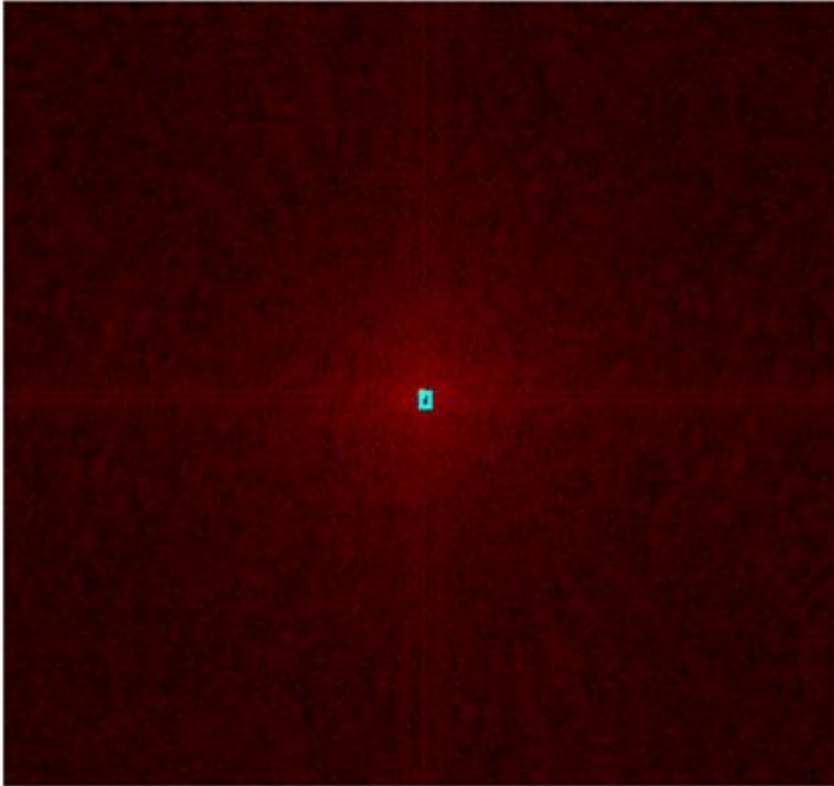
Reconstructed Image



What does the box do?

We can do the same thing in images!

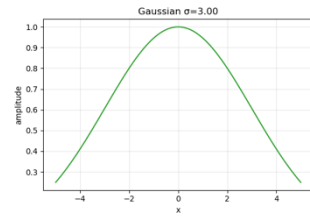
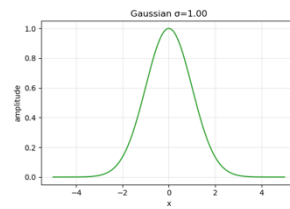
FFT with Mask (Size: 3px)



Reconstructed Image



Low-pass = Convolve with Gaussian



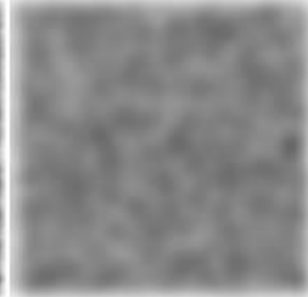
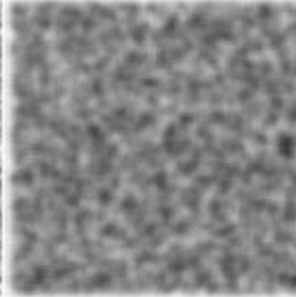
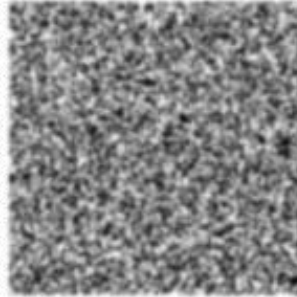
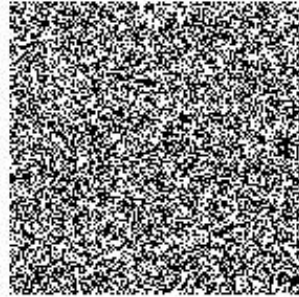
Gaussian blur radius:

1 px

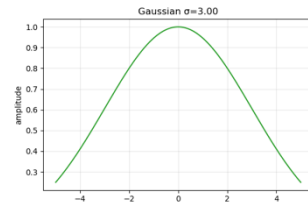
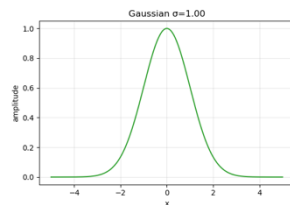
2 px

4 px

Image



Convolution theorem: convolving = multiplying frequencies



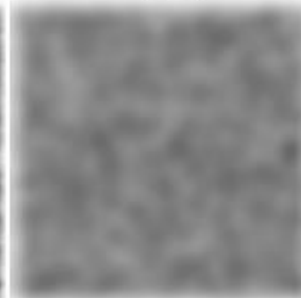
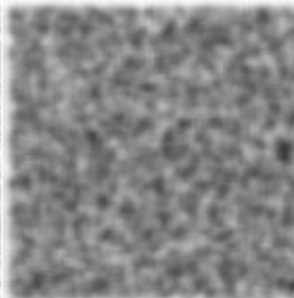
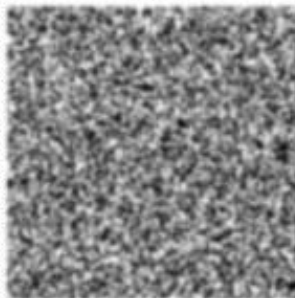
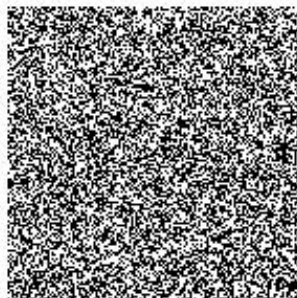
Gaussian blur radius:

1 px

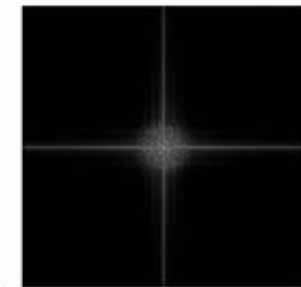
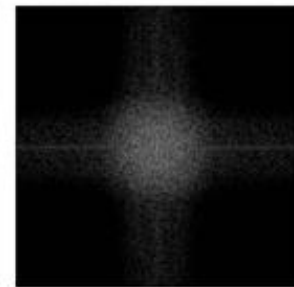
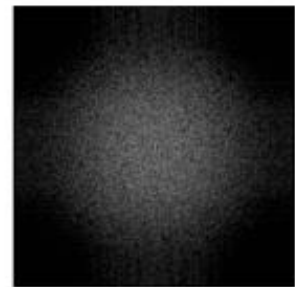
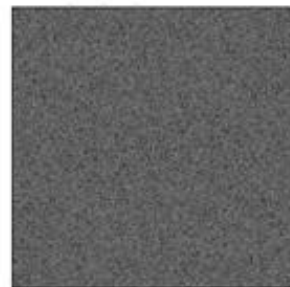
2 px

4 px

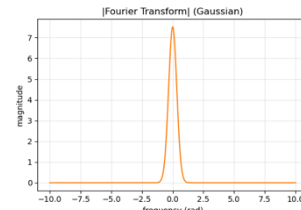
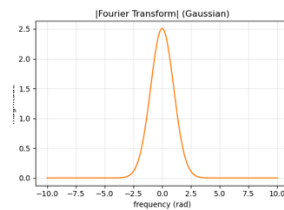
Image



Frequency
Image



X



Matching images with frequencies

A



B



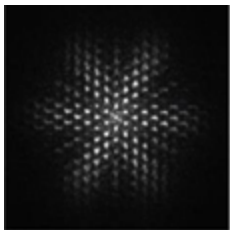
C



D



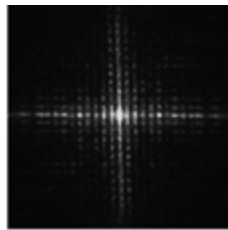
E



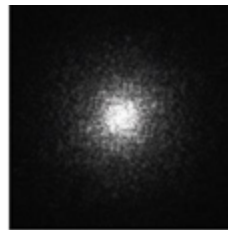
I



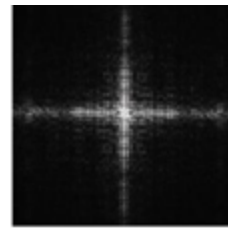
II



III



IV



V

Matching images with frequencies

A



B



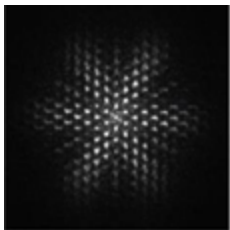
C



D



E



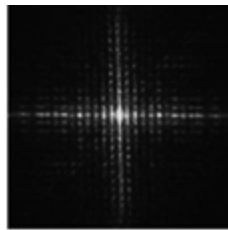
I

C



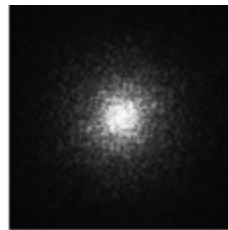
II

D



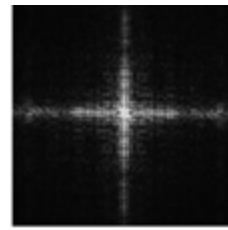
III

B



IV

E



V

A

Demo!

<https://cal-cs180.github.io/fa25/fft-tool.html>

3b1b has great visuals if you want to dive deeper!

